

C737

Evolution

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A teacher describes a beach mouse population on light sand dunes, studied since 1990 by a university team. Mice vary in fur color, and fur color is inherited. Owls catch more dark mice. A student says, "The mice turned light because they needed to hide." Which response best corrects the student?

- A. Owls caused mutations for light fur.
- B. The university team bred light mice to protect them.
- C. Light mice already existed; they survived and reproduced more, so light fur became more common over generations.
- D. Each mouse gradually lightened its fur to match the sand.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which change would NOT be passed on by natural selection?

- A. A gene variant for lactose tolerance in adults
- B. Any trait that helps an animal survive
- C. Beak depth in finches, which is strongly inherited
- D. A weightlifter's large muscles built through training

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

In a herd tracked by rangers using GPS collars, one large buck lives 12 years and fathers 2 surviving fawns. A smaller buck lives 6 years and fathers 8 surviving fawns. The herd grazes on a 3,000-acre preserve. Which buck has the higher evolutionary fitness?

- A. They are equal because both reproduced.
- B. The larger buck, because it lived twice as long
- C. The larger buck, because it is stronger
- D. The smaller buck

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A storm wipes out most of a wildflower population on a small island that a local hiking club visits each summer. By chance, the 20 surviving plants carry mostly white-flower alleles, although purple and white flowers had done equally well before. What process best explains the change in allele frequency?

- A. Mutation to white alleles caused by the storm
- B. Genetic drift (a bottleneck)
- C. Natural selection for white flowers
- D. Gene flow

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

Four populations of the same beetle species differ only in size. In which one is genetic drift most likely to change allele frequencies quickly?

- A. A population of 250,000
- B. The one with the most mutations
- C. A population of 25
- D. All four equally, since drift is random

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

Two frog populations in ponds 2 km apart, one of them next to a school, had become different in skin color. A new drainage ditch now lets frogs move between the ponds and breed. The ditch was dug by a county crew. What is the most likely long-term effect?

- A. The populations become more different.
- B. The populations become more similar genetically.
- C. No change, because only a few frogs move.
- D. They immediately become two separate species.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

Two cricket populations, recorded by a teacher on a phone app that logs sound, look identical. One group's males chirp at a pulse rate that the other group's females ignore, so the groups do not mate. Which type of barrier is this?

- A. Geographic (habitat) isolation
- B. Prezygotic (behavioral) isolation
- C. Postzygotic isolation
- D. No barrier, because they look the same.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

A horse and a donkey mate and produce a healthy mule, but mules are almost always sterile. What kind of barrier keeps horses and donkeys separate species?

- A. Prezygotic isolation
- B. No barrier, since they produced offspring
- C. Temporal isolation
- D. Postzygotic isolation (hybrid sterility)

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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9.

A museum exhibit that opened last year shows a bat wing, a human arm, a bird wing, and a butterfly wing. The exhibit has 40 panels. Which pair shows homologous structures?

- A. Bird wing and butterfly wing
- B. Bat wing and butterfly wing, because both are used to fly
- C. Bat wing and human arm
- D. None, because they all look different.

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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10.

A tree drawn on a whiteboard in bright markers shows, left to right, the tips lamprey, shark, frog, lizard, and bird. Lizard and bird split from their most recent common ancestor after frogs branched off. A student says frogs are closer to lizards than birds are, because frog sits next to lizard. Which statement is correct?

- A. Birds are more closely related to lizards than frogs are.
- B. Frogs, lizards, and birds are all equally related.
- C. Frogs are closer to lizards because the tips are adjacent.
- D. Lampreys are the ancestors of birds.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Species A and B share a common ancestor 5 million years ago. Species A and C share a common ancestor 20 million years ago. Which conclusion follows?

- A. A is more closely related to B than to C.
- B. C is four times older than B.
- C. B evolved from A.
- D. A is more closely related to C.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

A clinic in a town of 30,000 sees an infection that once cleared up with a common antibiotic but now often does not. Local news covered the story twice. What is the best evolutionary explanation?

- A. A few bacteria already had resistance; the drug killed the others, and the resistant ones multiplied.
- B. Patients' bodies became used to the antibiotic.
- C. Media coverage led people to stop taking it.
- D. The antibiotic caused the bacteria to develop resistance.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A teacher describes a beach mouse population on light sand dunes, ~~studied since 1990 by a university team.~~ Mice vary in fur color, and fur color is inherited. Owls catch more dark mice. A student says, "The mice turned light because they needed to hide." Which response best corrects the student?

Target word: correct explanation of adaptation **Decides:** Heritable variation plus differential survival.

A	Inverted operation	Predators select among variants; they do not cause them.
B	Camouflage	The research detail does not explain the change.
C	Correct	Selection acts on existing heritable variation across generations.
D	Confusion set	Individuals do not evolve; this is the inheritance-of-acquired-traits idea.

2. Answer D Concept 1 • transfer

Which change would NOT be passed on by natural selection?

Target word: reverse: identify a non-heritable change **Decides:** Only heritable traits respond to selection.

A	Inverted operation	Heritable, and has been selected in some human groups.
B	Absolute language	Helpful but non-heritable traits are not passed on.
C	Confusion set	Heritable variation is exactly what selection acts on.
D	Correct	Acquired during life, not inherited.

3. Answer D Concept 2

In a herd ~~tracked by rangers using GPS collars,~~ one large buck lives 12 years and fathers 2 surviving fawns. A smaller buck lives 6 years and fathers 8 surviving fawns. ~~The herd grazes on a 3,000-acre preserve.~~ Which buck has the higher evolutionary fitness?

Target word: compare fitness **Decides:** Surviving offspring: 8 vs. 2.

A	Dropped element	Ignores the difference in offspring number.
B	Adjacent category	Survival counts only as a path to offspring.
C	Confusion set	Strength matters only through reproduction.
D	Correct	More surviving offspring means higher fitness.

4. Answer B Concept 3

A storm wipes out most of a wildflower population on a small island ~~that a local hiking club visits each summer.~~ By chance, the 20 surviving plants carry mostly white-flower alleles, although purple and white flowers had done equally well before. What process best explains the change in allele frequency?

Target word: name the mechanism **Decides:** Random survival unrelated to the trait, in a small population.

A	Inverted operation	The storm removed plants; it did not create alleles.
B	Correct	Chance events reshaped allele frequencies.
C	Confusion set	Survival was random with respect to flower color.
D	Adjacent category	No alleles moved in from another population.

5. Answer C Concept 3 • transfer

Four populations of the same beetle species differ only in size. In which one is genetic drift most likely to change allele frequencies quickly?

Target word: where drift is strongest **Decides:** Drift is strongest in small populations.

A	Inverted operation	Large populations buffer chance events.
B	Adjacent category	Mutation is a separate mechanism.
C	Correct	Chance has the largest effect in the smallest population.
D	Absolute language	Random does not mean equal in effect across sizes.

6. Answer B Concept 4

Two frog populations in ponds 2 km apart, ~~one of them next to a school~~, had become different in skin color. A new drainage ditch now lets frogs move between the ponds and breed. ~~The ditch was dug by a county crew.~~ What is the most likely long-term effect?

Target word: effect of migration **Decides:** Gene flow mixes alleles.

A	Inverted operation	Mixing alleles works against divergence.
B	Correct	Gene flow reduces differences between populations.
C	Absolute language	Even low migration rates can move alleles over time.
D	Confusion set	Interbreeding works against speciation.

7. Answer B Concept 5

Two cricket populations, ~~recorded by a teacher on a phone app that logs sound~~, look identical. One group's males chirp at a pulse rate that the other group's females ignore, so the groups do not mate. Which type of barrier is this?

Target word: classify the barrier **Decides:** Mating never happens: a behavioral barrier before fertilization.

A	Adjacent category	The stem gives a signal difference, not a place difference.
B	Correct	Mating is blocked before any zygote forms.
C	Inverted operation	Postzygotic barriers act after fertilization.
D	Camouflage	Appearance does not decide whether they mate.

8. Answer D Concept 5 • transfer

A horse and a donkey mate and produce a healthy mule, but mules are almost always sterile. What kind of barrier keeps horses and donkeys separate species?

Target word: classify from the outcome **Decides:** A zygote forms; the hybrid cannot reproduce.

A	Inverted operation	They did mate and fertilization occurred.
B	Dropped element	Ignores that the offspring cannot reproduce.
C	Confusion set	A prezygotic barrier based on breeding season.
D	Correct	The barrier acts after fertilization.

9. Answer C Concept 6

A museum exhibit ~~that opened last year~~ shows a bat wing, a human arm, a bird wing, and a butterfly wing. ~~The exhibit has 40 panels.~~ Which pair shows homologous structures?

Target word: homologous pair **Decides:** Shared bone pattern from a common ancestor.

A	Confusion set	Analogous: same function, separate origins.
B	Adjacent category	Shared function points to analogy, not homology.
C	Correct	Same bones (humerus, radius, ulna) inherited from a shared ancestor.
D	Absolute language	Homology is about shared structure and ancestry, not appearance.

10. Answer A Concept 7

A tree ~~drawn on a whiteboard in bright markers~~ shows, left to right, the tips lamprey, shark, frog, lizard, and bird. Lizard and bird split from their most recent common ancestor after frogs branched off. A student says frogs are closer to lizards than birds are, because frog sits next to lizard. Which statement is correct?

Target word: relatedness on a tree **Decides:** Lizard and bird share a more recent common ancestor than either shares with frog.

A	Correct	The lizard–bird node is more recent than the node that includes frogs.
B	Dropped element	Ignores the branching order.
C	Confusion set	Tip order is arbitrary; nodes set relatedness.
D	Adjacent category	Living tips are relatives, not ancestors.

11. Answer A Concept 7 • transfer

Species A and B share a common ancestor 5 million years ago. Species A and C share a common ancestor 20 million years ago. Which conclusion follows?

Target word: relatedness from ancestor dates **Decides:** The more recent shared ancestor means closer relatives.

A	Correct	A and B share the more recent ancestor.
B	Wrong stem number	Split dates describe branch points, not the ages of living species.
C	Adjacent category	Sister species share an ancestor; one did not come from the other.
D	Inverted operation	An older common ancestor means more distant relatives.

12. Answer A Concept 8

A clinic ~~in a town of 30,000~~ sees an infection that once cleared up with a common antibiotic but now often does not. ~~Local news covered the story twice.~~ What is the best evolutionary explanation?

Target word: explain resistance **Decides:** The drug selects existing resistant variants.

A	Correct	Selection on pre-existing variation.
B	Confusion set	Resistance is in the bacteria, not the patients.
C	Camouflage	The news detail does not explain the genetic change in bacteria.
D	Inverted operation	The drug selects variants; it does not create them on purpose.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	9	
Absolute language	Always, never, guarantee — where judgment applies.	4	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Natural selection needs heritable variation	Selection changes a population when individuals vary in a heritable trait that affects survival and reproduction; individuals do not evolve, populations do.	Thinking organisms change because they need to, or pass on traits acquired during life.	“Adapted because it needed to,” or a trait gained during an individual’s lifetime.
2	Evolutionary fitness	Fitness is relative reproductive success: the number of surviving offspring passed to the next generation.	Equating fitness with strength, size, or lifespan.	Individuals compared by size or age alongside offspring counts.
3	Genetic drift	Drift is random change in allele frequencies, strongest in small populations and after bottlenecks or founder events.	Assuming every allele change is adaptive, or that drift matters in large populations as much as small.	A small or newly founded population, or a random disaster.
4	Gene flow	Migration that moves alleles between populations makes them more similar and can counter local adaptation.	Thinking migration always increases differences between populations.	Individuals moving between groups and breeding.
5	Reproductive isolation and speciation	Prezygotic barriers (habitat, timing, behavior, mechanics) block mating or fertilization; postzygotic barriers (hybrid death or sterility) act after a zygote forms.	Calling any barrier “postzygotic,” or treating hybrid sterility as preventing mating.	Different mating calls or seasons vs. sterile or weak hybrids.
6	Homologous vs. analogous structures	Homologous structures share ancestry even if functions differ; analogous structures share function from convergent evolution, not common ancestry.	Judging relatedness by similar function.	Wings, fins, or limbs compared across distant groups.
7	Reading phylogenetic trees	Relatedness is set by the most recent common ancestor (the node), not by how close the tips sit on the page.	Reading neighbors along the tips as closest relatives, or reading living species as ancestors of each other.	A tree and “most closely related.”
8	Evolution of drug resistance	Antibiotics do not create resistance mutations; they select for resistant variants that already exist, which then multiply.	Thinking bacteria “learn” or are caused by the drug to become resistant.	A population exposed to a drug that later stops working.

Course scope (public description): This course addresses why evolution is the fundamental concept that underlies all life sciences and how it contributes to advances in medicine, public health, and conservation. This course helps participants gain a firm understanding of the basic mechanisms of evolution including the process of speciation and how these systems have given rise to the great diversity of life in the world today. This course also explore how new ideas, discoveries, and technologies are modifying prior evolutionary concepts. Ultimately, the course will explain how evolution works and how we know what we know.